Plan

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1 0. Daily Protocol (New Requirement)

*Do this for 15 minutes **before** starting the day's problems.*

- ☐ **Theory Flashcards**: Recite the **exact statements** (assumptions & conclusions) of:
  - Bolzano-Weierstrass Theorem
  - Mean Value Theorem (MVT)
  - Intermediate Value Theorem (IVT)
  - Definition of Continuity ($\epsilon - \delta$)
  - **Definition of** π (Source: $\text{prediction}_{\text{ar}}$)

2 1. The "Red Alert" List (Highest Priority)

These are the "Must-Pass" items. I have mapped them to the exact problems in '$ps_{total.org}$'.

2.1 Sequences & Recursion

- **Recursion Limits (Certain):** ps_{04}, Problem 8.
  - **Task:** Prove bounded ($x_n \geq \sqrt{2}$), monotonic (decreasing), and find limit ($x = \frac{1}{2}(x + 2/x)$).
- **Series Convergence Criteria:** ps_{01}, Problem 3.
  - **Task:** Alternating Harmonic Series convergence.
- **Limits Evaluating to 0:** ps_{00}, Problem 3.
  - **Task:** Show $\lim \frac{(-1)^n}{n} = 0$.

2.2 Continuity

- **Composition of Functions (Classic):** ps_{02}, Problem 2.
  - **Task:** Prove continuity of $\frac{1}{x^2+1}$ and $\exp(|x|)$.
- **Euclidean Metric (Very High Relevance):** ps_{01}, Problem 2.
  - **Task:** Convergence in $\mathbb{R}^2$ with $d(x, y) = |x_1 - y_1| + |x_2 - y_2|$.
- **Continuity by Cases:** ps_{02}, Problem 3.
  - **Task:** Prove continuity at $x = 0$ for the piecewise function defined by $\exp(-1/x)$.

2.3 Functions & Differentiation

- **CRITICAL FAIL CONDITION (Derivatives):** ps_{04}, Problem 6.
  - **Task:** You must be able to differentiate e^{3x+1}, $\ln(5x^2...)$, and $\sqrt{1+e^{2x}}$ without error.
- **The "Promised" Problem (**exp(log) **trick):** ps_{04}, Problem 6 (General Skill).
  - **Task:** Practice derivatives of type $f(x)^{g(x)}$ (e.g., x^x) by rewriting as $\exp(g(x)\ln f(x))$.
- **Mean Value Theorem:** ps_{05}, Problem 2.
  - **Task:** Find ξ for $f(x) = \ln(x)$ on $[1, e]$.
- **Truncated Series:** ps_{03}, Problem 6.
  - **Task:** Write approximations for sin and cos.
- **Differentiation by Definition:** ps_{04}, Problem 3.
  - **Task:** Prove product rule differentiability at x_0 where $f(x_0) = 0$.

2.4 Integration

- **Improper Integrals:** ps_{07}, Problem 3.
  - **Task:** Determine s for which $\int_0^1 x^s dx$ is well-defined.
- **Integration Rules:** ps_{06}, Problems 3 & 4.
  - **Task:** Integration by parts (xe^x) and Substitution ($2xe^{x^2}$).
- **Sup/Inf Sums:** ps_{06}, Problem 1.
  - **Task:** Calculate $L(f, \mathcal{P}_n)$ and $U(f, \mathcal{P}_n)$ for $f(x) = 3x + 2$.

3 2. The Action Plan (7-Day Intensive Schedule)

3.1 Day 1: Foundations & Metric Spaces

Theme: "Mastering the Definition of Distance"

- ☐ **Theory Drill**: Definition of Metric, Triangle Inequality.
- ☒ **ps_{01}, Problem 1**: Discrete metric proofs.
- ☒ **ps_{01}, Problem 2**: Convergence in $\mathbb{R}^2$ (Euclidean vs. Taxi-cab).
- ☒ **ps_{00}, Problem 3**: Basic limits ($\epsilon - N$).

3.2 Day 2: Sequences, Recursion & Induction

Theme: "The Algorithm of Convergence"

- ☐ **Theory Drill**: Bolzano-Weierstrass, Monotone Convergence Thm.
- ☒ **$\mathbf{ps_{04}}$, Problem 8 (THE BIG ONE)**: Recursive sequence convergence.
- ☐ **$\mathbf{ps_{01}}$, Problem 4**: Euler number sequence $(1 + \frac{1}{n+1})^{n+1}$.
- ☐ **$\mathbf{ps_{04}}$, Problem 5 (Induction)**: [NEW]
  - Prove $\frac{d}{dx}x^n = nx^{n-1}$.
  - **Why:** High probability "easy points" question.
- ☒ **$\mathbf{ps_{01}}$, Problem 3**: Alternating Harmonic Series.
- ☒ **$\mathbf{ps_{00}}$, Problem 3**: Limits at 0.

3.3 Day 3: Continuity

Theme: "Breaking and Connecting Functions"

- ☐ **Theory Drill**: Def of Continuity, Intermediate Value Theorem.
- ☒ **$\mathbf{ps_{02}}$, Problem 2**: Continuity of Composition $(1/(x^2 + 1))$.
- ☒ **$\mathbf{ps_{02}}$, Problem 3**: Continuity by Cases (The $e^{-1/x}$ problem).
  - **Tip:** Check Left Limit vs Right Limit at $x = 0$.
- ☒ **$\mathbf{ps_{02}}$, Problem 1**: Dirichlet-style function.
- ☒ **$\mathbf{ps_{03}}$, Problem 1**: ReLU Function (Left/Right limits).

3.4 Day 4: Exponentials, Logarithms & Interpolation

Theme: "Constructing and Approximating"

- ☐ **Theory Drill**: Definition of $\exp(x)$ (Series vs Limit).
- ☒ **$\mathbf{ps_{03}}$, Problem 6**: Truncated Series $(\sin, \cos)$.
- ☒ **$\mathbf{ps_{03}}$, Problem 4**: Linear Approximation of $e^{1.3}$.
- ☒ **$\mathbf{ps_{03}}$, Problem 2**: Softplus activation $g(x) = \log(1 + e^x)$.
- ☒ **$\mathbf{ps_{02}}$, Problem 5**: Bisection Method / IVT Application.

3.5 Day 5: Differentiation & Mean Value Theorem

Theme: "Slopes and Inequalities"

- ☐ **Theory Drill**: Mean Value Theorem, Rolle's Theorem.
- ☒ **$\mathbf{ps_{05}}$, Problem 2 (MVT)**: Find ξ for $\ln(x)$ on $[1, e]$.
- ☐ **$\mathbf{ps_{04}}$, Problem 9 (Newton)**: [NEW]
  - **Task:** Apply the formula $x_{new} = x_{old} - f(x)/f'(x)$.
- ☒ **$\mathbf{ps_{04}}$, Problem 6 (Drill)**: Derivatives of $\exp(\log f(x))$.
- ☒ **$\mathbf{ps_{04}}$, Problem 3**: Differentiability Proof (Product Rule).
- ☒ **$\mathbf{ps_{05}}$, Problem 5**: Optimization (Maximize Area).

3.6 Day 6: Integration & Rules

Theme: "Summing it Up"

- ☐ **Theory Drill**: Fundamental Theorem of Calculus.
- ☒ **$\mathbf{ps_{06}}$, Problem 1**: Calculate Upper/Lower Sums explicitly.
- ☒ **$\mathbf{ps_{06}}$, Problem 3**: Integration by Parts.
- ☒ **$\mathbf{ps_{06}}$, Problem 4**: Substitution Rule.

3.7 Day 7: Improper Integrals, Taylor & Final Review

Theme: "The End Game"

- ☐ **Theory Drill**: Definition of π, Taylor's Theorem.
- ☐ **$\mathbf{ps_{07}}$, Problem 3**: Improper Integrals (Convergence of x^s).
- ☐ **$\mathbf{ps_{07}}$, Problem 4**: Series Convergence (Integral Test logic).
  - **Context:** Touches on the "Uniform Convergence" / Series topic flagged in predictions.
- ☐ **$\mathbf{ps_{07}}$, Problem 2**: Taylor Polynomials ($n = 2$).
- ☐ **Mock Exam**: Pick one random problem from Days 2, 3, and 5 to solve without notes.